

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

JULY 2023

Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

QP No. : UF009-A

Duration: 3 hours

Max. Marks: 100

PART-A

Answer all the questions

10 x 2 = 20 Marks

1. Expand $2\cos^2\theta - \cos^3\theta$ in ascending powers of θ as far as term involving θ^4 .
2. Evaluate i^i .
3. Expand $\sinh^5(\theta)$ in a series of hyperbolic sines of multiples of θ .
4. Find the index, signature and rank of the Quadratic form $x^2 + 2y^2 - 3z^2$.
5. Prove that A and A^T have the same eigenvalues.
6. Express the $X_1 = (1,2)$ as a linear combination of $X_2 = (2,3)$ and $X_3 = (4,5)$.
7. If α, β, γ are roots $x^3 - 3x + 1 = 0$, find the equation whose roots are $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$.
8. Find the cubic equation two of whose roots are $1 - i$ and 5.
9. Evaluate $\int_1^b \int_1^a \frac{dx dy}{xy}$.

10. Prove symmetry of the Beta function.

PART -B

Answer all the questions

4 x 15 = 60 Marks

11. If $x = 2\cos\theta$, show that $\frac{1+\cos 9\theta}{1+\cos\theta} = (x^4 - x^3 - 3x^2 + 2x + 1)^2$

(OR)

12. (a) If $\sin(A + iB) = x + iy$, prove that

$$i) \frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = 1 \text{ and } ii) \frac{x^2}{\sin^2 A} + \frac{y^2}{\cos^2 A} = 1 \quad (8)$$

(b) Prove that $\log\left(\frac{a+ib}{a-ib}\right) = 2i \tan^{-1}\left(\frac{b}{a}\right)$ and hence evaluate

$$\cos\left[i \log\left(\frac{a+ib}{a-ib}\right)\right] \quad (7)$$

13. Find the values of μ , for which the equations $x + 2y + z = 4$, $2x - y - z = 3\mu$ and $4x - 7y - 5z = \mu^2$ have a solution. For these values of μ , find the solutions also.

(OR)

14. Reduce the quadratic form into canonical form and also determine the nature of the quadratic form

$$2x^2 + y^2 - 3z^2 + 12xy - 8yz - 4zx.$$

15. (a) Solve the equation $x^4 - 15x^3 + 70x^2 - 120x + 64 = 0$ whose roots are in G.P. (8)

(b) Diminish the roots of the equation

$$x^4 - 12x^3 + 48x^2 - 72x + 35 = 0. \quad (7)$$

(OR)

16. (a) Solve the equation $x^6 + 2x^5 + 2x^4 - 2x^2 - 2x - 1 = 0$. (8)

- (b) Discuss the stability of the system represented by the characteristic equation $x^5 - 2x^4 - 10x^3 + 8x^2 + 33x + 18 = 0$, using Routh array method. (7)

17. (a) Find the area of astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ using gamma functions. (8)
(b) Evaluate $\int_C xy \, ds$, where C is the arc of the parabola $y^2 = 4x$ between the vertex and the positive end of the latus rectum. (7)

(OR)

18. (a) Prove that $\int_0^\infty \sqrt{x} e^{-x^8} dx * \int_0^\infty \frac{e^{-x^4}}{\sqrt{x}} dx = \frac{\pi}{16\sqrt{2}}$ (8)
(b) Change the order of integration in $\int_0^a \int_0^{\frac{b\sqrt{a^2-x^2}}{a}} x^2 dy \, dx$ and then integrate it. (7)

PART-C

Answer the following

1 x 20 = 20 Marks

19. Find the SVD of the matrix $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$.
